

SHANKER VALIPIREDDY

Principal Architect — Applied AI & Forward Deployment — Agentic AI, GenAI, LLMs

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PROFESSIONAL SUMMARY

Principal Architect and applied-AI leader with **20+ years** of experience taking AI systems from discovery through pilot, stabilization, and **production rollout** in enterprise environments. Currently Principal Architect / AI Enablement Lead at **Aflac**, driving enterprise **GenAI and agentic AI** adoption across **100+ engineering teams** — acting as the main liaison between AI teams and business teams, delivering targeted workshops and standardized AI playbooks that moved teams from code-completion tools to agentic environments. Hands-on Python builder of multi-agent systems (LangGraph, CrewAI, MCP, LangChain agents), enterprise LLM evaluation and governance (Human-in-the-Loop guardrails, defect-escape and token-efficiency measurement), and cloud-scale deployment (AWS, Azure, GCP, Kubernetes). Trusted advisor to VPs and directors, with a track record of mentorship, cross-functional delivery, and production-grade AI engineering.

CORE SKILLS

Forward Deployment & Applied AI: discovery-to-production rollout, customer-embedded delivery, reference implementations, reusable AI playbooks, stakeholder communication, ambiguity navigation, customer support chatbots

Agentic AI & LLMs: multi-agent systems (LangGraph, CrewAI, LangChain Agents, **MCP**), LLM APIs (Azure OpenAI API, Claude, GPT-4, GPT, Llama 2/3.1, Mistral, Cohere, Anthropic), **RAG** (LlamaIndex), prompt engineering, fine-tuning, LLM structured output, LangSmith, LangServe

Insurance AI & Claims Automation: claims text extraction, **claims automation**, claims processing, claims adjudication, **touchless automation**, document digitization, business rules encoding (policy/processing manuals), claims turnaround & productivity improvement

Document Processing & NLP: OCR (**AWS Textract**), document processing, **entity extraction**, text classification, sentiment analysis, NLP (BERT, RoBERTa), knowledge models, graph databases (Neptune DB), document management (FileNet)

ML & MLOps: machine learning, model training, model deployment, feature engineering, TensorFlow, Keras, **AWS SageMaker**, **MLOps**, scalable AI pipelines

Data Engineering: ETL (InfoWorks), EMR clusters, AWS S3/EC2/Lambda/Step Functions, **Kafka** real-time processing, **Snowflake**, DB2, API Gateways, AWS Databricks

Engineering: Python, Java 17, TypeScript, **React/Next.js**, Tailwind CSS, **FastAPI**, Flask API, Streamlit, Vercel, Jupyter Notebooks, Spring Boot, SQL, REST APIs

Deployment & Platforms: AWS, Azure, GCP (Vertex AI), EKS, **Kubernetes**, Docker, CI/CD (Azure DevOps), **inference cost/latency/perf optimization**

Evaluation & Safety: **LLM evaluation** (token efficiency, defect-escape rates, QA pass rates), **Human-in-the-Loop guardrails**, feedback loops and agent self-improvement, AI governance

Leadership: cross-functional team leadership (**15+ onshore/offshore**), **agile methodologies**, sprint planning, cross-training, offshore support, mentoring and coaching, executive briefings

WORK EXPERIENCE

Principal Architect / AI Enablement Lead — AI/ML, Agentic AI, Gen AI | Aflac – GA 01.2025 – Present

- Spearheaded enterprise AI enablement program: drove **GenAI and agentic AI** adoption across **100+ engineering teams** by delivering targeted, role-specific workshops and standardized '**AI Playbooks**' transitioning teams from code-completion tools (Copilot) to agentic environments (Kiro, Cursor, Devin).
- Led end-to-end deployment of a **large-scale AI modernization**: orchestrated migration of a legacy \$3M/year ETL platform to cloud-native **AWS** (Airflow, Glue), using **AI agents** to refactor complex pipelines and modernize Java/Python enterprise applications.
- Deployed an **agentic code-generation pipeline** using **AWS Kiro agents** that automated conversion of **~20k data pipelines** across 82 business domains into Airflow DAGs and Glue PySpark jobs — achieving a **99.3% QA pass rate** with structured evaluation.
- Implemented **AI governance** and performance metrics: measured AI adoption and ROI via Jira cycle time, code-defect escape rates, and token efficiency; implemented **Human-in-the-Loop guardrails** for code quality, security, and compliance in **production**.
- Designed and built a **Human-in-the-Loop review application** (Angular 17 + Spring Boot 3.x) for domain experts to approve/reject/retire converted pipelines, with structured feedback loops driving **agent self-improvement**.
- Established **CI/CD** with integrated security (Azure DevOps, Cloudsmith, Veracode scanning, SonarQube); built knowledge models on **graph databases (Neptune DB)** to encode **business rules** and processing manuals.
- Reported to **VP Innovation & Strategy** as main liaison between the AI team and business teams; managed a cross-functional team of **15+ onshore/offshore** resources with sprint-based capacity planning and cross-training.

Principal Architect – AI/ML, Agentic AI, Gen AI (Full Time) | Brains Technology Solutions – MI
01.2023 – 01.2025

- Designed and delivered complex **multi-agent systems** using **LangGraph/CrewAI, MCP, LangChain** agents and **LLM structured outputs** — systems that reason, take actions, and interact with external tools, improving the functionality and interactivity of LLM applications.
- Developed and implemented **Generative AI models** using **GPT, Claude, Llama, Mistral, BERT**; built proof-of-concepts with **Agentic AI and LLMs** (GCP, Vertex AI) for regulatory text extraction as part of JPMC RCMA (regulatory change management automation).

- Built LLM development and **fine-tuning** workflows (Llama 2/3.1, Mistral, Cohere, Anthropic/Claude, GPT-4) with **MLOps** practices for **model training, evaluation, deployment**, and feature engineering (LangGraph, LangChain, LlamaIndex, LangSmith, TensorFlow, Keras).
- Applied **transformer NLP** (RoBERTa, DistilBERT) for text analytics and automation; designed AI/ML infrastructure to run models on **TensorFlow, PyTorch, NumPy**; integrated AI systems with **React/Next.js, TypeScript, FastAPI/Flask**, SQL front ends.
- Delivered **customer-facing AI proof-of-concepts** and pilots, translating ambiguous business problems into working LLM systems and **production-ready deployment** patterns.

Senior Principal / Technologist – AI/ML, Big Data (Full Time) | Western Digital – CA 01.2016 – 01.2023

- Designed and delivered an **AI/ML data platform** on composable infrastructure and **Kubernetes**, delivered as on-demand **SaaS** for edge-to-cloud AI use cases (video surveillance, cyber threat intelligence, 360° video streaming, image/video analytics); demonstrated price/performance benefits across industry verticals.
- Delivered key components of a **data lakehouse** (MongoDB, Cassandra, Delta Lake) and **AI/ML pipelines**; delivered optimal Percona MySQL solutions on zoned storage; prototyped smart archiving and analytics on OpenStack, significantly reducing defect detection times.
- Published patents and white papers, delivered conference sessions, and mentored/coached engineering teams for professional growth and successful deliveries.
- Drove **inference** and data-pipeline **performance optimization** across **GPUs, RDMA/RoCE, NVMe-oF** platforms, computational storage, and object storage — translating workload findings into architectural recommendations.

Principal Architect – Big Data, AI/ML, NLP | Cisco Systems, Inc. – CA (Contract) 10.2014 – 12.2015

- Built **deep-learning anomaly detection** over network-device telemetry: converted log text to vectors with **pre-trained embeddings**, trained models to understand log semantics and context, and applied clustering to flag anomalies.
- Integrated the system with live devices for **real-time anomaly detection** and built an operator dashboard; implemented a human feedback mechanism so administrators could confirm/dismiss anomalies and the model refined itself over time.

Sr. Java Developer | Franklin Templeton – CA 10.2010 – 10.2014

- Built **Hadoop/HBase ETL** and MapReduce analytics; Hive/Pig reporting and Oracle integration via Sqoop; delivered Twitter **sentiment analysis** via HBase REST API for business users.

Java Engineer | IBM, WIPRO and CF India 06.2003 – 10.2010

- Enterprise application integration (Java/Spring, TIBCO); proof-of-concept design and development.

EDUCATION

B.Tech, Computer Science and Engineering (Jan 1999 – Jun 2003)

PATENTS

Method and Apparatus for Smart Archiving and Analytics — US 10,360,193 (Issued Jul 23, 2019)

Intelligent Data Access Across Tiered Storage Systems — US 11,544,216 (Issued Jan 3, 2023)

"Enabling an Optimal Application Ecosystem on ZNS Storage" — Flash Memory Summit, August 2022, Santa Clara, CA

"An Analytics System on OpenStack™ for Manufacturing" — Corporate White Paper, 2018, Western Digital

"An Analytics System for Optimizing Manufacturing Processes" — Corporate White Paper, 2018, Western Digital

"Use Cases for ActiveScale™ Data Pipeline Service" — Corporate White Paper, 2019, Western Digital

"ActiveScale™ Data Pipeline Service with Elasticsearch — Best Practices for Metadata Indexing and Search" — Corporate White Paper, 2019, Western Digital

"Flash Storage Complementing a Data Lake for Real-time Insight" — Flash Memory Summit 2018, Santa Clara, CA

"AI Solution — Object Storage Content Analysis" — Corporate Presentation, 2018, Western Digital

"AI/ML MLPerf/TensorFlow/PyTorch Performance Benchmark — Image Detection" — Corporate Presentation, 2018, Western Digital

"AI/ML/Deep Learning (TensorFlow/Keras) for Analyzing Fraud Detection" — Presentation, 2019

"An Artificial Intelligence Data Pipeline for Storing and Processing Ingested Data" — Flash Memory Summit 2019, Santa Clara, CA

PUBLICATIONS & CONFERENCE TALKS

CERTIFICATIONS

AWS Certified Machine Learning - Specialty (AWS, Sep 2024) • AZ-102 Microsoft Azure AI Engineer: Solution Design (Microsoft, Jun 2024) • Generative AI with Large Language Models (Coursera) • Machine Learning Specialization (Coursera) • Natural Language Processing Specialization (Coursera) • Natural Language Processing with Attention Models (Coursera) • NLP with Classification and Vector Spaces (Coursera) • Neural Networks and Deep Learning (Coursera) • Transformer Models and BERT (Google) • Large Language Models: Application through Production (edX) • Building Conversational AI Applications (NVIDIA, Oct 2023) • Introduction to Responsible AI (Google, Jun 2023) • Introduction to AI in the Data Center (NVIDIA, Dec 2019) • Global Artificial Intelligence Bootcamp (Feb 2021) • Monitoring Kubernetes with Prometheus and Grafana (Coursera) • Smart Contracts - Ethereum (Coursera, Aug 2018) • Spark Certificate for Scalable Machine Learning (edX, Aug 2015) • Big Data XSeries, UC BerkeleyX (edX) • Apache Cassandra Certified Developer (DataStax, Sep 2015) • MongoDB Certified Developer (Apr 2013) • Advanced Predictive/Data Analytics using R (IIT Hyderabad, Jun 2014) • PMP Training (Star PMO, Jan 2023)

AWARDS

Best Employee award